Hall Effect in a Doped Germanium Semiconductor

Determination of the Hall Coefficient, Carrier Concentration, and Carrier Mobility

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September 1st, 2026

Abstract

The Hall effect was studied in lightly doped p-type and n-type germanium crystals using a Holmarc Hall Effect Apparatus. The electromagnet was calibrated against coil current, and the Hall voltage was measured as a function of sample current at two fixed fields ($B_1 = 1570$ G, $B_2 = 3010$ G) and two sample temperatures (33.7°C, 45°C), for both carrier types. The zero-field longitudinal voltage was subtracted from each field-on reading to isolate the true Hall signal, and the slope of the resulting linear region gave V_H/I_H at each condition. At 33.7°C, the p-type sample gave $R_H = (7588.7 \pm 21.1)$ cm^{3}C^{-1}, $n = 8.236 \times 10^{14}$ cm^{-3}, $\mu = 758.9 \pm 2.1$ cm^{2}V^{-1}s^{-1}, with close agreement between the two fields. The n-type sample at the same temperature gave $R_H = (-7748.8 \pm 424.0)$ cm^{3}C^{-1}, $n = -8.066 \times 10^{14}$ cm^{-3}, $\mu = -774.9 \pm 42.4$ cm^{2}V^{-1}s^{-1}. At 45°C both samples showed larger scatter between the two fields, consistent with increased thermal noise and self-heating at elevated temperature. The sign of R_H correctly identified holes as the majority carrier in the p-type sample and electrons in the n-type sample at both temperatures.

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1 Introduction

When a current-carrying conductor is placed in a magnetic field perpendicular to the current, the moving charge carriers experience a Lorentz force that deflects them toward one edge of the sample. This builds up a transverse electric field that, in steady state, balances the Lorentz force, producing a measurable transverse voltage known as the Hall voltage. This phenomenon, the Hall effect, is a direct probe of the sign, density, and mobility of the charge carriers in a conducting or semiconducting material.

Semiconductors exhibit the Hall effect far more strongly than metals because their carrier densities are several orders of magnitude lower. Doped (extrinsic) semiconductors are therefore the standard material for demonstrating and quantifying the effect. In this experiment, lightly doped p-type and n-type germanium samples were studied. The aims were:

- To calibrate the electromagnet used to generate the transverse field.
- To measure the Hall voltage as a function of sample current in each sample, at two field strengths and two sample temperatures.
- To determine the Hall coefficient R_H, majority carrier density n, and carrier mobility μ for both samples, and to identify the majority carrier type from the sign of R_H.

2 Theory

Consider a rectangular semiconducting slab of thickness d carrying current I_H along its length, placed in a magnetic field B perpendicular to the current direction. The Lorentz force on the charge carriers deflects them toward one face of the sample, producing a transverse Hall voltage V_H given by

$$V_H = \frac{1}{ne} \cdot \frac{BI_H}{d}, \tag{1}$$

where n is the carrier concentration and $e = 1.6 \times 10^{-19}$ C is the electronic charge. The quantity $R_H = 1/(ne)$ is the Hall coefficient. Rearranging Eq. (1),

$$R_H = \frac{V_H}{I_H} \cdot \frac{d}{B}. \tag{2}$$

The sign of R_H identifies the majority carrier: negative for electrons (n-type), positive for holes (p-type). Carrier concentration follows from

$$n = \frac{1}{R_H\, e}. \tag{3}$$

Carrier mobility μ is obtained by combining R_H with the sample conductivity $\sigma = 1/\rho$,

$$\mu = R_H\, \sigma. \tag{4}$$

Experimentally, V_H/I_H is the slope of a linear fit to V_H against I_H at fixed B, after subtracting the zero-field ($B = 0$) voltage at each current value to remove the ohmic offset caused by any longitudinal misalignment of the Hall contacts:

$$V_{H,\mathrm{net}}(I_H) = V_{\mathrm{measured}}(I_H,\, B \neq 0) - V_{\mathrm{measured}}(I_H,\, B = 0). \tag{5}$$

Because d and B are quoted in the apparatus manual in mm and Gauss while V_H/I_H is in mV/mA, Eq. (2) is written with an explicit unit-conversion factor 10^8,

$$R_H = \frac{V_H}{I_H} \cdot \frac{d}{B} \times 10^8 \ \ \mathrm{cm^3 C^{-1}}, \qquad d \text{ in cm},\ B \text{ in Gauss}. \tag{6}$$

3 Experimental Setup and Procedure

3.1 Apparatus

The measurements were performed using a Holmarc Hall Effect Apparatus (Model HO-ED-EM-06), consisting of the following components:

- **Hall Effect Control Unit** – supplies and displays the sample current (0–10 mA), Hall voltage (0–200.0 mV), and sample temperature (ambient–100°C) on a $3\frac{1}{2}$-digit LCD display.
- **Hall Effect cartridge** – a lightly doped germanium (Ge) crystal of thickness $d = 0.5$ mm and resistivity $\rho = 10\ \Omega \cdot \text{cm}$, mounted with an integrated heating element and temperature sensor.
- **Electromagnet** – provides a variable magnetic field of 0–5000 $\pm$ 5% Gauss across a 20 mm air gap, driven by a Constant Current Power Supply (0–3.5 A).
- **Digital Gauss Meter** – an Indium Arsenide (InAs) Hall probe transducer, range 0–5 kG, accuracy $\pm$0.5%, used to calibrate the electromagnet.

3.2 Procedure

1. The Gauss probe was mounted at the centre of the pole-piece air gap and zeroed. The magnet current was increased from 0 to 3.0 A in 0.5 A steps, and the flux density was read twice at each step and averaged (Table 1).
2. With no field applied, the longitudinal voltage was recorded for each sample as the sample current was increased from 0 to 7.0 mA, at sample temperatures of 33.7°C and 45°C.
3. The magnet current was set to 1 A (measured $B_1 = 1570$ Gauss) and the Hall voltage was recorded over the same current range, for both samples and both temperatures.
4. Step 3 was repeated with the magnet current at 2 A (measured $B_2 = 3010$ Gauss).
5. The net Hall voltage, Eq. (5), was obtained by subtracting the Step 2 baseline from the Steps 3–4 readings at each matching sample current.

3.3 Sources of Uncertainty

The dominant sources of uncertainty were:
(i) the $\pm$0.5% accuracy of the digital Gauss meter;
(ii) sample self-heating at higher currents, causing visible curvature above roughly 4.5 mA (p-type) and 1.5–2 mA (n-type), mitigated by restricting fits to the pre-saturation region;
(iii) the nominal sample thickness and resistivity, taken as manufacturer-specified values rather than independently measured, which propagate directly into R_H, n, and μ; and
(iv) increased scatter at 45°C, where the heater-controlled temperature and the sample's own resistive self-heating both add noise to the voltage readings.

4 Observations and Data

Table 1: Calibration of magnetic flux density.

I (A)	B_1 (×10 G)	B_2 (×10 G)	Avg. B (×10 G)
0.0	1	1	1.0
0.5	68	85	76.5
1.0	147	167	157.0
1.5	222	241	231.5
2.0	293	309	301.0
2.5	359	370	364.5
3.0	424	424	424.0

Table 2: Longitudinal voltage at $B = 0$, all recorded points, both samples and temperatures.

I (mA)	V_p @33.7°C (mV)	V_p @45°C (mV)	V_n @33.7°C (mV)	V_n @45°C (mV)
0.0	2.3	1.8	−1.4	−1.2
0.5	3.6	4.0	−1.8	−1.6
1.0	5.5	6.1	−2.3	−2.1
1.5	7.8	8.2	−3.1	−2.8
2.0	11.0	11.7	−3.9	−3.8
2.5	14.4	15.3	–	−4.7
3.0	16.9	17.5	–	−5.3
3.5	18.6	19.8	–	–
4.0	20.7	21.7	–	–
4.5	22.5	23.6	–	–
5.0	24.7	26.2	–	–
5.5	25.3	29.4	–	–
6.0	25.2	29.5	–	–
6.5	–	30.4	–	–
7.0	–	30.3	–	–

The subscripts n and p denote the type of semiconductor. These values are subtracted, point by point at matching I_H, from the field-on Hall voltages of Table 3 to remove the ohmic offset caused by any longitudinal misalignment of the Hall contacts. Where a current value in Table 3 has no matching entry here (as the baseline sweep did not extend that far), that current is simply not available for net–Hall-voltage computation and is marked '–'.

For p-type and n-type semiconductors, at magnetic fields $B_1 = 1570$ Gauss & $B_2 = 3010$ Gauss, and temperatures $T_1 \approx 33.7$ °C & $T_2 \approx 45.0$ °C:

Table 3: Raw field-on Hall voltage

	p-type, raw V_H (mV)				n-type, raw V_H (mV)			
I (mA)	B_1,33.7	B_1,45	B_2,33.7	B_2,45	B_1,33.7	B_1,45	B_2,33.7	B_2,45
0.0	–	–	–	–	−3.1	−2.8	−4.7	−3.6
0.5	5.9	6.6	7.8	8.1	−4.2	−3.7	−6.3	−5.0
1.0	8.9	9.3	11.4	11.5	−5.6	−4.8	−8.5	−6.9
1.5	12.0	13.3	16.2	16.7	−7.4	−6.2	−11.7	−9.7
2.0	17.1	18.3	22.7	23.3	−10.3	−7.9	−15.8	−14.2
2.5	22.6	24.2	29.9	30.1	−10.6	−9.2	−16.6	−17.8
3.0	25.5	28.5	34.3	34.6	–	−10.8	–	−20.5
3.5	28.8	31.5	37.7	39.3	–	−12.4	–	−20.8
4.0	31.8	34.8	41.5	42.5	–	−12.9	–	−20.9
4.5	34.4	37.9	45.3	46.8	–	−13.0	–	−21.2
5.0	38.2	41.4	49.7	51.4	–	−13.2	–	−21.3
5.5	39.7	44.8	53.5	53.8	–	–	–	–
6.0	39.9	45.1	53.5	54.7	–	–	–	–
6.5	–	44.7	–	–	–	–	–	–

Table 4: Net Hall voltage $V_{H,\text{net}} = V_{H,\text{raw}} - V(B = 0)$

	p-type, net V_H (mV)				n-type, net V_H (mV)			
I (mA)	B_1,33.7	B_1,45	B_2,33.7	B_2,45	B_1,33.7	B_1,45	B_2,33.7	B_2,45
0.0	–	–	–	–	−1.70	−1.60	−3.30	−2.40
0.5	2.30	4.80	4.20	6.30	−2.40	−2.10	−4.50	−3.40
1.0	3.40	5.30	5.90	7.50	−3.30	−2.70	−6.20	−4.80
1.5	4.20	7.20	8.40	10.60	−4.30	−3.40	−8.60	−6.90
2.0	6.10	10.10	11.70	15.10	−6.40	−4.10	−11.90	−10.40
2.5	8.20	12.50	15.50	18.40	−6.60	−4.50	–	−13.10
3.0	8.60	13.20	17.40	19.30	–	−5.50	–	−15.20
3.5	10.20	14.00	19.10	21.80	–	–	–	–
4.0	11.10	15.00	20.80	22.70	–	–	–	–
4.5	11.90	16.20	22.80	25.10	–	–	–	–
5.0	13.50	17.80	25.00	27.80	–	–	–	–
5.5	14.40	18.60	28.20	27.60	–	–	–	–
6.0	14.70	15.70	28.30	25.30	–	–	–	–
6.5	–	15.20	–	–	–	–	–	–

Points which showed visible saturation were excluded from the linear fits. The n-type zero-field baseline at 33.7°C (Table 2) was itself recorded only up to $I_H = 2.0$ mA, removing saturation points.

5 Analysis and Results

5.1 Plotting

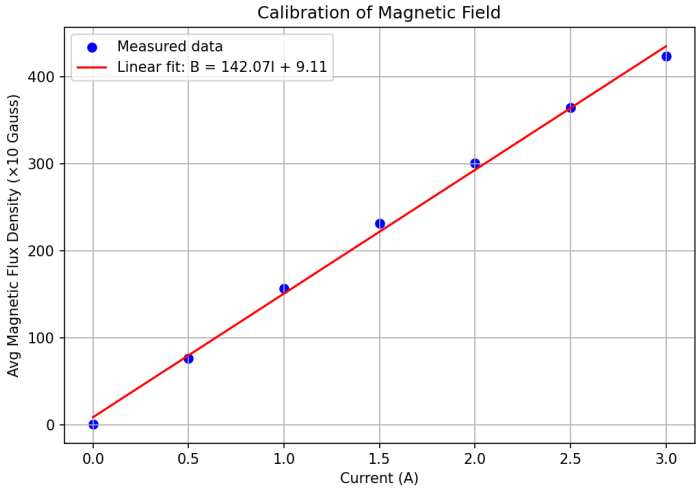


Figure 1: Calibration of the electromagnet: magnetic flux density B versus coil current I, with best-fit line $B = (1420.7 \pm 90.4)\, I + (91.1 \pm 168.5)$ Gauss ($R^2 = 0.997$).

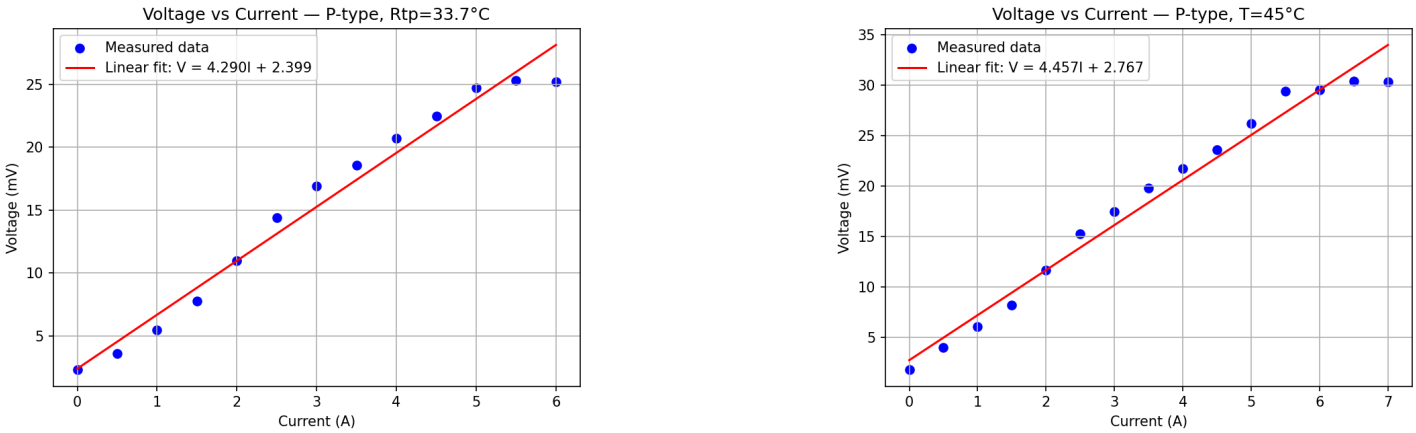


Figure 2: Zero-field voltage vs Current for P-type sample at different temperatures

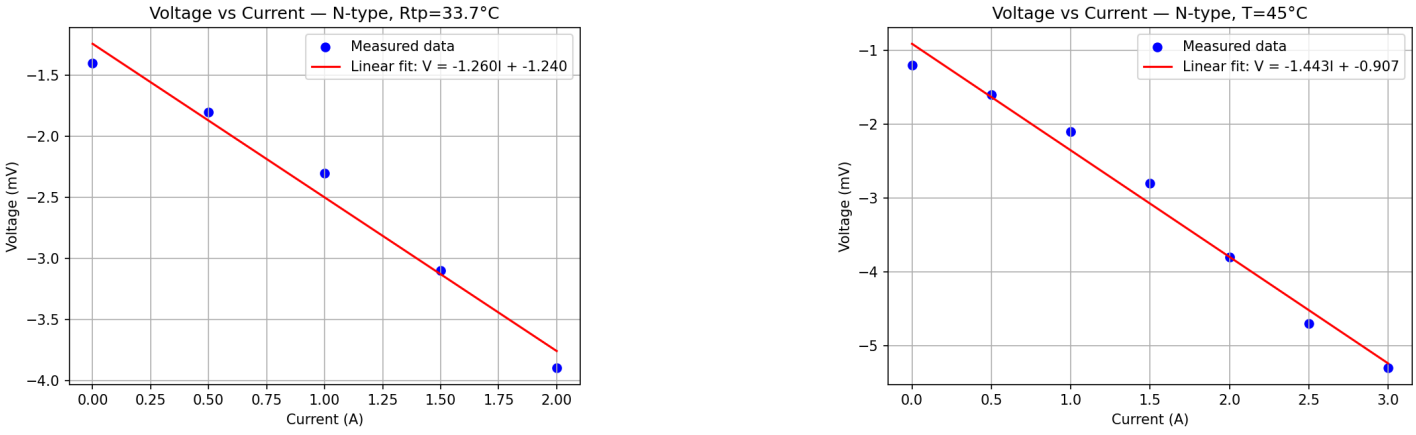
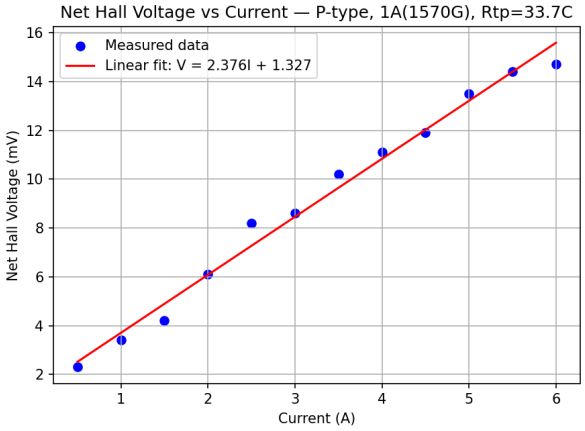
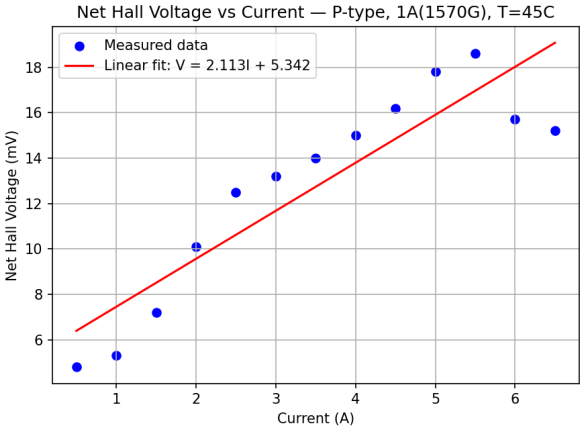


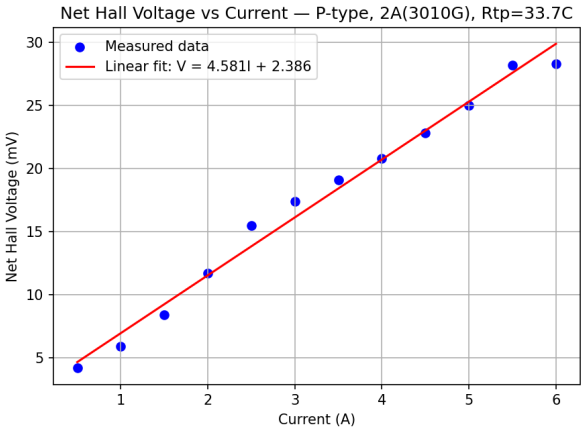
Figure 3: Voltage vs Current for N-type sample at different temperatures

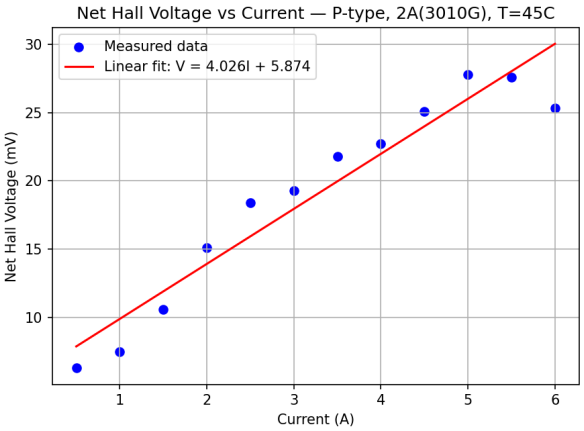
Least square fit plots for background-subtracted V_H versus I_H

1) P-type semiconductor

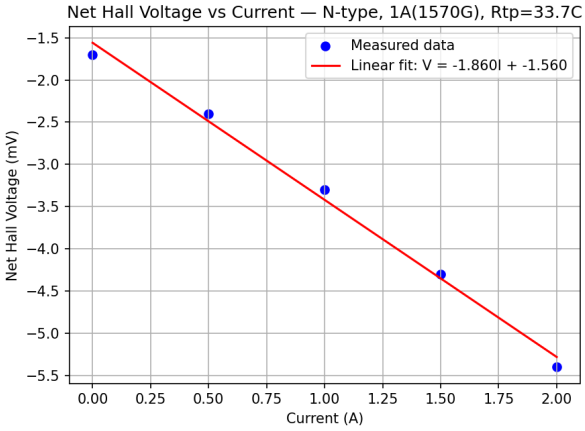


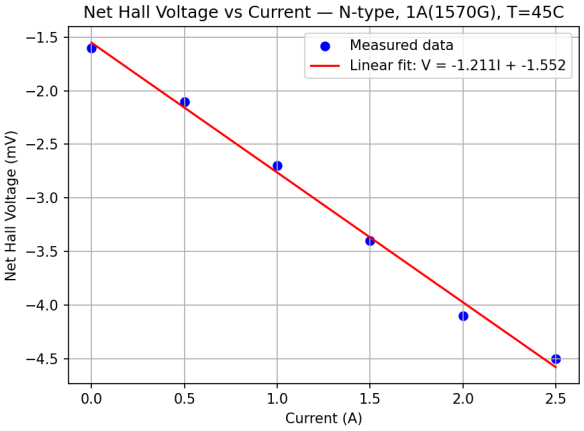


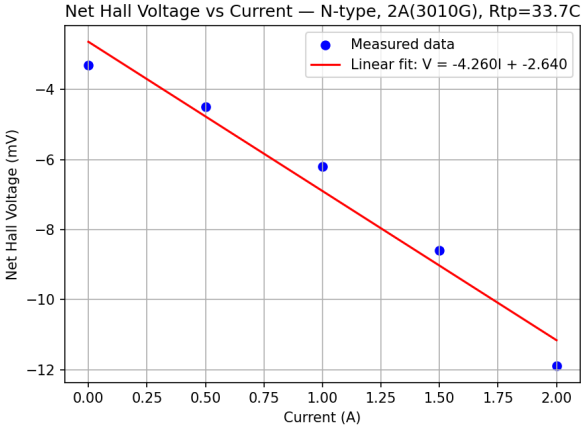


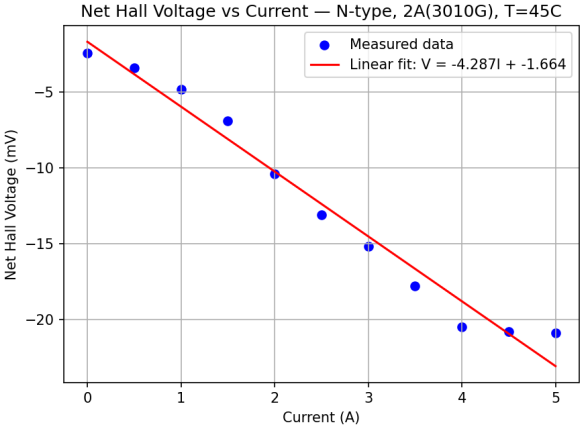


2) N-type semiconductor









5.2 Calibration Fit

A linear least-squares fit of average B (in units of $\times 10$ Gauss) against I over all 7 points gives

$$B = m_B I + c_B, \qquad m_B = 142.07, \quad c_B = 9.11, \qquad R^2 = 0.997.$$

Calculation of m_B (least squares): with $\bar{I} = 1.5$, $\bar{B} = 165.07$ ($\times 10$ G),

$$m_B = \frac{\sum (I_i - \bar{I})(B_i - \bar{B})}{\sum (I_i - \bar{I})^2} = 142.07 \ (\times 10 \text{ G/A}).$$

Verifying the two field settings used in the Hall measurements:

$$B_1 = (142.07)(1) + 9.11 \approx 157 \ (\times 10 \text{ G}) = 1570 \text{ G},$$

$$B_2 = (142.07)(2) + 9.11 \approx 301 \ (\times 10 \text{ G}) = 3010 \text{ G}.$$

5.3 Slopes of the Net Hall Voltage

For each of the eight net-Hall-voltage columns in Table 4, a least-squares line $V_{H,\text{net}} = m\, I_H + c$ was fitted using every available point in that column. The resulting slopes, intercepts, and coefficients of determination are given in Table 5.

Table 5: Linear fit parameters for $V_{H,\text{net}}$ vs. I_H, using all available points per column.

Sample	Condition	N pts	Slope m (mV/mA)	Intercept c (mV)	R^2
p-type	33.7°C, $B_1 = 1570$ G	12	2.3762	1.3273	0.9865
p-type	33.7°C, $B_2 = 3010$ G	12	4.5811	2.3864	0.9869
p-type	45°C, $B_1 = 1570$ G	13	2.1132	5.3423	0.8135
p-type	45°C, $B_2 = 3010$ G	12	4.0259	5.8742	0.9203
n-type	33.7°C, $B_1 = 1570$ G	5	−2.3000	−1.1500	0.9542
n-type	33.7°C, $B_2 = 3010$ G	4	−4.9200	−1.6500	0.9792
n-type	45°C, $B_1 = 1570$ G	6	−1.3200	−1.4067	0.9915
n-type	45°C, $B_2 = 3010$ G	6	−4.9943	−0.2267	0.9865

5.4 Hall Coefficient, Carrier Density and Mobility

Using Eq. (6) with $d = 0.05$ cm, $\sigma = 1/\rho = 0.1\,\Omega^{-1}\text{cm}^{-1}$, and $e = 1.6 \times 10^{-19}$ C, each slope in Table 5 was converted to R_H, then to n (Eq. (3)) and μ (Eq. (4)). Full results are collected in Table 6.

Worked example (p-type, 33.7°C, $B_1 = 1570$ G). From Table 5, $m = V_H/I_H = 2.3762$ mV/mA. Applying Eq. (2) with $d = 0.05$ cm:

$$R_H = 2.3762 \times \frac{0.05}{1570} \times 10^8 = 2.3762 \times 3.1847 \times 10^{-5} \times 10^8 = 7567.5 \text{ cm}^3\text{C}^{-1}.$$

Then, from Eq. (3) and (4),

$$n = \frac{1}{R_H e} = \frac{1}{(7567.5)(1.6 \times 10^{-19})} = 8.259 \times 10^{14} \text{ cm}^{-3},$$

$$\mu = R_H\, \sigma = (7567.5)(0.1) = 756.75 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}.$$

For each sample at each temperature, the two field-dependent values of R_H (at B_1 and B_2) are averaged; the uncertainty is taken as the standard error of this two-point average, $\text{SE} = s/\sqrt{2}$ where s is the sample standard deviation of the pair.

p-type, 33.7°C:

$$\bar{R}_H = \frac{7567.52 + 7609.80}{2} = 7588.66 \text{ cm}^3\text{C}^{-1}, \qquad \text{SE} = 21.14 \text{ cm}^3\text{C}^{-1}.$$

$$\bar{n} = \frac{1}{\bar{R}_H e} = 8.236 \times 10^{14} \text{ cm}^{-3}, \qquad \bar{\mu} = \bar{R}_H \sigma = 758.87 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}.$$

p-type, 45°C:

$$\bar{R}_H = \frac{6729.94 + 6687.54}{2} = 6708.74 \text{ cm}^3\text{C}^{-1}, \qquad \text{SE} = 21.20 \text{ cm}^3\text{C}^{-1}.$$

$$\bar{n} = 9.316 \times 10^{14} \text{ cm}^{-3}, \qquad \bar{\mu} = 670.87 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}.$$

n-type, 33.7°C:

$$\bar{R}_H = \frac{-7324.84 - 8172.76}{2} = -7748.80 \text{ cm}^3\text{C}^{-1}, \qquad \text{SE} = 423.96 \text{ cm}^3\text{C}^{-1}.$$

$$\bar{n} = -8.066 \times 10^{14} \text{ cm}^{-3}, \qquad \bar{\mu} = -774.88 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}.$$

n-type, 45°C:

$$\bar{R}_H = \frac{-4203.82 - 8296.18}{2} = -6250.00 \text{ cm}^3\text{C}^{-1}, \qquad \text{SE} = 2046.18 \text{ cm}^3\text{C}^{-1}.$$

$$\bar{n} = -1.000 \times 10^{15} \text{ cm}^{-3}, \qquad \bar{\mu} = -625.00 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}.$$

Table 6: Final averaged results.

Sample	Temperature	R_H (cm^3C^{-1})	n (cm^{-3})	μ ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$)
p-type	33.7°C	7588.66 ± 21.14	8.236×10^{14}	758.87 ± 2.11
p-type	45°C	6708.74 ± 21.20	9.316×10^{14}	670.87 ± 2.12
n-type	33.7°C	-7748.80 ± 423.96	-8.066×10^{14}	-774.88 ± 42.40
n-type	45°C	-6250.00 ± 2046.18	-1.000×10^{15}	-625.00 ± 204.62

6 Discussion

At 33.7°C, the two field settings gave R_H values agreeing to within 0.6% for the p-type sample and 10.9% for the n-type sample, indicating a reasonably reliable, field-independent Hall coefficient at this temperature. At 45°C, the agreement is markedly worse for n-type: the two fields differ by only 0.6% for p-type but by 65% for n-type, the latter reflected in the very large standard error on R_H for n-type at 45°C in Table 6. This is attributed to the combined effect of active heater-controlled temperature and resistive self-heating of the small germanium crystal at the higher sample currents required to resolve a measurable Hall signal.

In both samples and at both temperatures, the sign of R_H correctly identified the majority carrier: positive for the p-type sample (holes) and negative for the n-type sample (electrons). The carrier densities obtained, of order 10^{14}–10^{15} cm^{-3}, are pertaining to light extrinsic doping of germanium, several orders of magnitude below typical metallic carrier densities. The largest systematic uncertainty not captured in the quoted standard errors is the sample thickness and resistivity, both taken as nominal manufacturer values ($d = 0.5$ mm, $\rho = 10$ Ωcm) rather than independently measured; any deviation from these propagates linearly into every value of R_H, n, and μ reported here. Retaining 2 points of saturation (points visibly affected by self-heating at high current) inflated the scatter between the two field-based estimates of R_H for n-type at 45°C.

7 Conclusion

The Hall effect was characterized in p-type and n-type germanium using a calibrated electromagnet and a Holmarc apparatus, at two field strengths and two sample temperatures, with all recorded data points retained in every fit. At 33.7°C, $R_H = (7588.7 \pm 21.1)$ cm^{3}C^{-1} for p-type and $R_H = (-7748.8 \pm 424.0)$ cm^{3}C^{-1} for n-type, giving carrier densities $n_p = 8.236 \times 10^{14}$ cm^{-3} and $n_n = -8.066 \times 10^{14}$ cm^{-3}, and mobilities $\mu_p = 758.9 \pm 2.1$ cm^{2}V^{-1}s^{-1} and $\mu_n = -774.9 \pm 42.4$ cm^{2}V^{-1}s^{-1}. The sign of R_H correctly distinguished the majority carrier in each sample. Results at 45°C were directionally consistent ($R_H = 6708.7 \pm 21.2$ cm^{3}C^{-1} for p-type, $R_H = -6250.0 \pm 2046.2$ cm^{3}C^{-1} for n-type) but carried substantially larger uncertainty for n-type, attributable to thermal effects at elevated sample temperature. The stated aims of the experiment were achieved.

Acknowledgments

Data were recorded using the departmental Holmarc Hall Effect Apparatus (Model HO-ED-EM-06) at the Physics Laboratory, IISER Bhopal.